

Table 7: Well-learned digits / performance-preserving digits of models tested on NUPA Test according to exact match.

	Add Int	Add Float	Add Frac	Add Easy Frac	Add Sci	Sub Int	Sub Float	Sub Frac
GPT-4o-mini	5 / 20	4 / 11	0 / 1	0 / 2	0 / 0	6 / 20	5 / 15	0 / 1
GPT-4o	5 / 20	4 / 11	0 / 1	0 / 1	0 / 0	6 / 20	4 / 15	0 / 1
Qwen2-72B	6 / 20	0 / 15	0 / 1	0 / 1	0 / 11	6 / 20	0 / 15	0 / 1
Qwen2-7B	4 / 14	0 / 15	0 / 1	0 / 0	0 / 7	4 / 14	0 / 13	0 / 1
Llama-3.1-8B-ft	4 / 12	5 / 17	0 / 1	0 / 1	0 / 4	4 / 12	4 / 17	0 / 2
Llama-3.1-70B	4 / 15	0 / 11	0 / 1	0 / 1	0 / 0	6 / 11	4 / 11	0 / 1
Llama-3.1-8B	4 / 9	3 / 11	0 / 1	0 / 0	0 / 0	3 / 10	3 / 10	0 / 1
Mixtral-8x7B	5 / 10	4 / 11	0 / 1	0 / 1	0 / 6	4 / 15	3 / 11	0 / 1
Llama-2-7b-hf	0 / 6	0 / 7	0 / 0	0 / 0	0 / 0	0 / 6	0 / 5	0 / 1

	Sub Sci	Max Int	Max Float	Max Frac	Max Sci	Max Hard Int	Max Hard Float	Max Hard Sci
GPT-4o-mini	5 / 20	4 / 11	0 / 1	0 / 2	0 / 0	6 / 20	5 / 15	0 / 1
GPT-4o	5 / 20	4 / 11	0 / 1	0 / 1	0 / 0	6 / 20	4 / 15	0 / 1
Qwen2-72B	6 / 20	0 / 15	0 / 1	0 / 1	0 / 11	6 / 20	0 / 15	0 / 1
Qwen2-7B	4 / 14	0 / 15	0 / 1	0 / 0	0 / 7	4 / 14	0 / 13	0 / 1
Llama-3.1-8B-ft	4 / 12	5 / 17	0 / 1	0 / 1	0 / 4	4 / 12	4 / 17	0 / 2
Llama-3.1-70B	4 / 15	0 / 11	0 / 1	0 / 1	0 / 0	6 / 11	4 / 11	0 / 1
Llama-3.1-8B	4 / 9	3 / 11	0 / 1	0 / 0	0 / 0	3 / 10	3 / 10	0 / 1
Mixtral-8x7B	5 / 10	4 / 11	0 / 1	0 / 1	0 / 6	4 / 15	3 / 11	0 / 1
Llama-2-7b-hf	0 / 6	0 / 7	0 / 0	0 / 0	0 / 0	0 / 6	0 / 5	0 / 1

	Multiply Hard Int	Multiply Hard Float	Multiply Hard Frac	Multiply Hard Sci	Multiply Easy Int	Multiply Easy Float	Multiply Easy Frac	Multiply Easy Sci
GPT-4o-mini	5 / 20	4 / 11	0 / 1	0 / 2	0 / 0	6 / 20	5 / 15	0 / 1
GPT-4o	5 / 20	4 / 11	0 / 1	0 / 1	0 / 0	6 / 20	4 / 15	0 / 1
Qwen2-72B	6 / 20	0 / 15	0 / 1	0 / 1	0 / 11	6 / 20	0 / 15	0 / 1
Qwen2-7B	4 / 14	0 / 15	0 / 1	0 / 0	0 / 7	4 / 14	0 / 13	0 / 1
Llama-3.1-8B-ft	4 / 12	5 / 17	0 / 1	0 / 1	0 / 4	4 / 12	4 / 17	0 / 2
Llama-3.1-70B	4 / 15	0 / 11	0 / 1	0 / 1	0 / 0	6 / 11	4 / 11	0 / 1
Llama-3.1-8B	4 / 9	3 / 11	0 / 1	0 / 0	0 / 0	3 / 10	3 / 10	0 / 1
Mixtral-8x7B	5 / 10	4 / 11	0 / 1	0 / 1	0 / 6	4 / 15	3 / 11	0 / 1
Llama-2-7b-hf	0 / 6	0 / 7	0 / 0	0 / 0	0 / 0	0 / 6	0 / 5	0 / 1

	Digit Max Int	Digit Max Float	Digit Add Int	Digit Add Float	Get Digit Int	Get Digit Float	Length Int	Length Float
GPT-4o-mini	5 / 20	4 / 11	0 / 1	0 / 2	0 / 0	6 / 20	5 / 15	0 / 1
GPT-4o	5 / 20	4 / 11	0 / 1	0 / 1	0 / 0	6 / 20	4 / 15	0 / 1
Qwen2-72B	6 / 20	0 / 15	0 / 1	0 / 1	0 / 11	6 / 20	0 / 15	0 / 1
Qwen2-7B	4 / 14	0 / 15	0 / 1	0 / 0	0 / 7	4 / 14	0 / 13	0 / 1
Llama-3.1-8B-ft	4 / 12	5 / 17	0 / 1	0 / 1	0 / 4	4 / 12	4 / 17	0 / 2
Llama-3.1-70B	4 / 15	0 / 11	0 / 1	0 / 1	0 / 0	6 / 11	4 / 11	0 / 1
Llama-3.1-8B	4 / 9	3 / 11	0 / 1	0 / 0	0 / 0	3 / 10	3 / 10	0 / 1
Mixtral-8x7B	5 / 10	4 / 11	0 / 1	0 / 1	0 / 6	4 / 15	3 / 11	0 / 1
Llama-2-7b-hf	0 / 6	0 / 7	0 / 0	0 / 0	0 / 0	0 / 6	0 / 5	0 / 1

	Truediv Int	Truediv Frac	Floordiv Int	Mod Int	Mod Easy Int	To Float Frac	To Float Sci	To Scient Int
GPT-4o-mini	5 / 20	4 / 11	0 / 1	0 / 2	0 / 0	6 / 20	5 / 15	0 / 1
GPT-4o	5 / 20	4 / 11	0 / 1	0 / 1	0 / 0	6 / 20	4 / 15	0 / 1
Qwen2-72B	6 / 20	0 / 15	0 / 1	0 / 1	0 / 11	6 / 20	0 / 15	0 / 1
Qwen2-7B	4 / 14	0 / 15	0 / 1	0 / 0	0 / 7	4 / 14	0 / 13	0 / 1
Llama-3.1-8B-ft	4 / 12	5 / 17	0 / 1	0 / 1	0 / 4	4 / 12	4 / 17	0 / 2
Llama-3.1-70B	4 / 15	0 / 11	0 / 1	0 / 1	0 / 0	6 / 11	4 / 11	0 / 1
Llama-3.1-8B	4 / 9	3 / 11	0 / 1	0 / 0	0 / 0	3 / 10	3 / 10	0 / 1
Mixtral-8x7B	5 / 10	4 / 11	0 / 1	0 / 1	0 / 6	4 / 15	3 / 11	0 / 1
Llama-2-7b-hf	0 / 6	0 / 7	0 / 0	0 / 0	0 / 0	0 / 6	0 / 5	0 / 1

	To Scient Float	Count Int	Sig Int
GPT-4o-mini	5 / 20	4 / 11	0 / 1
GPT-4o	5 / 20	4 / 11	0 / 1
Qwen2-72B	6 / 20	0 / 15	0 / 1
Qwen2-7B	4 / 14	0 / 15	0 / 1
Llama-3.1-8B-ft	4 / 12	5 / 17	0 / 1
Llama-3.1-70B	4 / 15	0 / 11	0 / 1
Llama-3.1-8B	4 / 9	3 / 11	0 / 1
Mixtral-8x7B	5 / 10	4 / 11	0 / 1
Llama-2-7b-hf	0 / 6	0 / 7	0 / 0

Table 8: Well-learned digits / performance-preserving digits of models tested on NUPA Test according to digit match.

	Add Int	Add Float	Add Frac	Add Easy Frac	Add Sci	Sub Int	Sub Float	Sub Frac
GPT-4o-mini	9 / 20	6 / 20	0 / 2	0 / 2	0 / 14	11 / 20	9 / 20	0 / 1
GPT-4o	10 / 20	6 / 20	0 / 1	0 / 2	0 / 14	11 / 20	10 / 20	0 / 1
Qwen2-72B	7 / 16	7 / 20	0 / 0	0 / 0	0 / 20	6 / 15	0 / 16	0 / 0
Qwen2-7B	4 / 12	3 / 20	0 / 0	0 / 0	0 / 0	6 / 14	6 / 20	0 / 0
Llama-3.1-8B-ft	6 / 16	9 / 20	0 / 1	0 / 0	0 / 7	6 / 19	12 / 20	0 / 1
Llama-3.1-70B	6 / 15	7 / 20	0 / 1	0 / 1	0 / 4	6 / 17	7 / 20	0 / 1
Llama-3.1-8B	4 / 9	6 / 18	0 / 0	0 / 0	0 / 0	6 / 11	6 / 17	0 / 0
Mixtral-8x7B	6 / 12	5 / 18	0 / 1	0 / 1	0 / 6	5 / 15	5 / 18	0 / 0
Llama-2-7b-hf	3 / 6	0 / 8	0 / 0	0 / 0	0 / 0	3 / 5	0 / 9	0 / 0

	Sub Sci	Max Int	Max Float	Max Frac	Max Sci	Max Hard Int	Max Hard Float	Max Hard Sci
GPT-4o-mini	0 / 3	100 / 100	10 / 100	0 / 7	19 / 98	100 / 100	8 / 100	18 / 100
GPT-4o	0 / 0	100 / 100	10 / 100	0 / 7	19 / 98	100 / 100	8 / 100	16 / 100
Qwen2-72B	0 / 0	21 / 100	30 / 100	0 / 4	100 / 100	82 / 100	32 / 100	100 / 100
Qwen2-7B	0 / 0	11 / 86	10 / 98	0 / 0	0 / 82	13 / 100	7 / 100	6 / 69
Llama-3.1-8B-ft	0 / 0	83 / 100	75 / 100	0 / 20	100 / 100	100 / 100	79 / 100	100 / 100
Llama-3.1-70B	0 / 0	0 / 86	0 / 98	0 / 4	0 / 0	0 / 0	0 / 100	0 / 100
Llama-3.1-8B	0 / 0	0 / 67	0 / 93	0 / 0	0 / 54	0 / 100	0 / 96	0 / 100
Mixtral-8x7B	0 / 0	7 / 21	3 / 19	0 / 0	0 / 62	20 / 100	4 / 25	0 / 100
Llama-2-7b-hf	0 / 0	0 / 100	0 / 22	0 / 0	0 / 17	99 / 100	0 / 38	0 / 100

	Multiply Hard Int	Multiply Hard Float	Multiply Hard Frac	Multiply Hard Sci	Multiply Easy Int	Multiply Easy Float	Multiply Easy Frac	Multiply Easy Sci
GPT-4o-mini	0 / 5	0 / 0	1 / 2	0 / 4	0 / 6	0 / 0	1 / 2	0 / 4
GPT-4o	0 / 5	0 / 0	1 / 2	0 / 4	0 / 6	0 / 0	1 / 3	0 / 4
Qwen2-72B	0 / 5	0 / 0	0 / 0	0 / 0	0 / 6	0 / 0	0 / 0	0 / 0
Qwen2-7B	0 / 3	0 / 0	0 / 0	0 / 0	0 / 4	0 / 0	0 / 0	0 / 0
Llama-3.1-8B-ft	0 / 4	0 / 0	0 / 3	0 / 4	0 / 6	0 / 3	1 / 3	0 / 5
Llama-3.1-70B	0 / 5	0 / 3	0 / 2	0 / 3	0 / 6	0 / 3	0 / 2	0 / 3
Llama-3.1-8B	0 / 4	0 / 0	0 / 0	0 / 0	0 / 5	0 / 0	0 / 1	0 / 0
Mixtral-8x7B	0 / 4	0 / 0	0 / 1	0 / 0	0 / 5	0 / 0	0 / 2	0 / 0
Llama-2-7b-hf	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0

	Digit Max Int	Digit Max Float	Digit Add Int	Digit Add Float	Truediv Int	Truediv Frac	Floordiv Int	Mod Int
GPT-4o-mini	0 / 100	0 / 100	0 / 0	0 / 0	0 / 5	1 / 2	5 / 15	0 / 3
GPT-4o	0 / 100	0 / 100	0 / 0	0 / 0	0 / 6	1 / 2	5 / 15	0 / 3
Qwen2-72B	0 / 100	0 / 100	0 / 0	0 / 0	0 / 0	0 / 1	3 / 8	0 / 3
Qwen2-7B	0 / 100	0 / 33	0 / 5	0 / 8	0 / 0	0 / 1	3 / 6	0 / 3
Llama-3.1-8B-ft	0 / 100	13 / 100	0 / 20	5 / 73	3 / 20	0 / 1	6 / 18	0 / 3
Llama-3.1-70B	0 / 100	0 / 100	0 / 0	0 / 0	0 / 0	0 / 1	4 / 12	0 / 0
Llama-3.1-8B	0 / 92	0 / 0	0 / 0	0 / 0	0 / 0	0 / 0	3 / 7	0 / 0
Mixtral-8x7B	0 / 100	0 / 97	0 / 5	0 / 5	0 / 0	0 / 1	3 / 7	0 / 3
Llama-2-7b-hf	0 / 100	0 / 42	0 / 0	0 / 0	0 / 0	0 / 0	0 / 3	0 / 0

	Mod Easy Int	To Float Frac	To Float Sci	To Scient Int	To Scient Float	Sig Int
GPT-4o-mini	0 / 3	0 / 8	0 / 9	23 / 67	8 / 24	31 / 100
GPT-4o	0 / 3	0 / 8	0 / 9	23 / 71	8 / 28	31 / 100
Qwen2-72B	0 / 3	6 / 9	16 / 28	100 / 100	96 / 100	100 / 100
Qwen2-7B	0 / 3	4 / 8	6 / 22	100 / 100	0 / 100	19 / 100
Llama-3.1-8B-ft	0 / 3	3 / 8	10 / 36	95 / 100	25 / 83	19 / 100
Llama-3.1-70B	0 / 0	4 / 9	0 / 12	0 / 100	0 / 21	0 / 21
Llama-3.1-8B	0 / 0	3 / 6	0 / 12	0 / 0	0 / 0	0 / 0
Mixtral-8x7B	0 / 3	4 / 9	14 / 36	100 / 100	68 / 100	17 / 100
Llama-2-7b-hf	0 / 0	0 / 5	0 / 0	0 / 34	0 / 22	0 / 0

Table 9: Well-learned digits / performance-preserving digits of models tested on NUPA Test according to dlength.

	Add Int	Add Float	Add Frac	Add Easy Frac	Add Sci	Sub Int	Sub Float	Sub Frac
GPT-4o-mini	20 / 20	4 / 8	0 / 2	1 / 2	0 / 0	20 / 20	5 / 9	0 / 1
GPT-4o	20 / 20	4 / 7	0 / 2	1 / 2	0 / 0	20 / 20	5 / 10	0 / 1
Qwen2-72B	20 / 20	7 / 13	0 / 0	0 / 0	0 / 0	20 / 20	0 / 0	0 / 0
Qwen2-7B	11 / 12	0 / 13	0 / 0	0 / 0	0 / 0	10 / 12	0 / 12	0 / 0
Llama-3.1-8B-ft	19 / 20	11 / 20	0 / 7	0 / 5	0 / 7	20 / 20	13 / 20	0 / 5
Llama-3.1-70B	20 / 20	6 / 13	0 / 1	0 / 1	0 / 0	20 / 20	4 / 13	0 / 1
Llama-3.1-8B	11 / 20	4 / 11	0 / 1	0 / 0	0 / 0	10 / 20	6 / 10	0 / 1
Mixtral-8x7B	10 / 14	4 / 12	0 / 1	0 / 1	0 / 0	16 / 20	5 / 13	0 / 0
Llama-2-7b-hf	5 / 12	0 / 10	0 / 0	0 / 0	0 / 0	4 / 20	0 / 11	0 / 1
	Sub Sci	Max Int	Max Float	Max Frac	Max Sci	Max Hard Int	Max Hard Float	Max Hard Sci
GPT-4o-mini	0 / 0	39 / 98	6 / 9	0 / 4	13 / 17	100 / 100	6 / 8	6 / 16
GPT-4o	0 / 0	31 / 72	6 / 9	0 / 2	12 / 17	100 / 100	6 / 8	8 / 16
Qwen2-72B	0 / 0	18 / 32	13 / 30	0 / 0	43 / 86	68 / 85	16 / 26	35 / 63
Qwen2-7B	0 / 0	11 / 23	0 / 17	0 / 0	0 / 14	86 / 100	0 / 20	0 / 11
Llama-3.1-8B-ft	0 / 0	47 / 83	35 / 54	0 / 5	97 / 100	100 / 100	25 / 52	35 / 69
Llama-3.1-70B	0 / 0	0 / 5	0 / 13	0 / 0	0 / 5	0 / 8	0 / 13	0 / 4
Llama-3.1-8B	0 / 0	0 / 0	0 / 7	0 / 0	0 / 0	0 / 18	0 / 6	0 / 0
Mixtral-8x7B	0 / 0	7 / 10	3 / 9	0 / 0	0 / 5	20 / 28	3 / 11	0 / 6
Llama-2-7b-hf	0 / 0	0 / 10	0 / 0	0 / 0	0 / 0	100 / 100	0 / 0	0 / 11
	Multiply Hard Int	Multiply Hard Float	Multiply Hard Frac	Multiply Hard Sci	Multiply Easy Int	Multiply Easy Float	Multiply Easy Frac	Multiply Easy Sci
GPT-4o-mini	7 / 11	0 / 0	1 / 2	0 / 0	9 / 20	0 / 0	1 / 2	0 / 0
GPT-4o	7 / 11	0 / 0	1 / 2	0 / 0	16 / 20	0 / 0	1 / 2	0 / 0
Qwen2-72B	0 / 6	0 / 0	0 / 0	0 / 0	5 / 7	0 / 0	0 / 0	0 / 0
Qwen2-7B	0 / 0	0 / 0	0 / 0	0 / 0	0 / 5	0 / 0	0 / 0	0 / 0
Llama-3.1-8B-ft	11 / 18	3 / 9	0 / 11	0 / 7	16 / 19	8 / 17	1 / 15	0 / 20
Llama-3.1-70B	5 / 9	0 / 3	0 / 1	0 / 0	7 / 16	0 / 3	0 / 1	0 / 0
Llama-3.1-8B	0 / 4	0 / 0	0 / 0	0 / 0	0 / 8	0 / 0	0 / 1	0 / 0
Mixtral-8x7B	4 / 6	0 / 0	0 / 2	0 / 0	4 / 8	0 / 0	0 / 2	0 / 0
Llama-2-7b-hf	0 / 0	0 / 0	0 / 0	0 / 0	0 / 5	0 / 0	0 / 0	0 / 0
	Digit Max Int	Digit Max Float	Digit Add Int	Digit Add Float	Truediv Int	Truediv Frac	Floordiv Int	Mod Int
GPT-4o-mini	5 / 16	0 / 7	0 / 8	0 / 0	0 / 5	1 / 2	8 / 14	3 / 15
GPT-4o	10 / 16	3 / 7	0 / 8	0 / 0	0 / 5	1 / 2	8 / 14	3 / 15
Qwen2-72B	7 / 16	0 / 9	0 / 0	0 / 0	0 / 0	0 / 1	5 / 7	0 / 5
Qwen2-7B	6 / 11	0 / 4	0 / 20	0 / 9	0 / 0	0 / 1	4 / 6	0 / 5
Llama-3.1-8B-ft	23 / 61	19 / 29	26 / 61	29 / 45	3 / 20	0 / 3	9 / 18	0 / 15
Llama-3.1-70B	6 / 16	0 / 7	0 / 11	0 / 0	0 / 0	0 / 1	3 / 10	0 / 4
Llama-3.1-8B	0 / 6	0 / 0	0 / 10	0 / 7	0 / 0	0 / 0	4 / 8	0 / 5
Mixtral-8x7B	0 / 12	0 / 4	0 / 8	0 / 7	0 / 0	0 / 1	4 / 7	0 / 5
Llama-2-7b-hf	0 / 7	0 / 3	0 / 9	0 / 7	0 / 0	0 / 1	0 / 3	0 / 6
	Mod Easy Int	To Float Frac	To Float Sci	To Scient Int	To Scient Float	Sig Int		
GPT-4o-mini	3 / 20	0 / 4	0 / 15	9 / 21	5 / 8	0 / 23		
GPT-4o	3 / 20	0 / 4	0 / 15	9 / 21	3 / 8	0 / 23		
Qwen2-72B	0 / 10	3 / 8	12 / 46	0 / 44	0 / 0	6 / 19		
Qwen2-7B	0 / 7	3 / 7	76 / 93	0 / 0	0 / 0	0 / 5		
Llama-3.1-8B-ft	0 / 17	3 / 9	14 / 78	27 / 33	16 / 38	0 / 0		
Llama-3.1-70B	0 / 5	3 / 7	0 / 15	0 / 0	0 / 0	0 / 0		
Llama-3.1-8B	0 / 6	3 / 6	0 / 12	0 / 0	0 / 0	0 / 0		
Mixtral-8x7B	0 / 6	3 / 6	15 / 16	0 / 18	0 / 0	0 / 0		
Llama-2-7b-hf	0 / 6	0 / 4	93 / 98	0 / 0	0 / 0	0 / 0		

Table 14: 8-digit digit match accuracy with small model (1.3M) with RoPE, NoPE and Alibi.

	Int-add	Float-add	Fraction-multiplication	Scientific-add
RoPE	0.091	0.88	0.23	0.75
NoPE	0.061	0.39	0.17	0.52
Alibi	0.056	0.31	0.18	0.50